

THE GRIM TEST

Granularity-Related
Inconsistency of
Means — detecting
impossible reported
averages in studies
with integer response
scales

At a glance: The GRIM test checks whether a reported mean is arithmetically possible given the sample size and the fact that the underlying data are integers. If no combination of integer responses could produce the reported mean, the paper contains an error — or something worse.

1. Overview

When survey or experimental data are collected on integer response scales — a 7-point Likert scale, a count of correct answers, a yes/no score converted to 0/1 — every data point is a whole number. The mean of a set of integers is constrained: not every decimal value is reachable given the number of observations. The GRIM test exploits this constraint.

Introduced by Nicholas Brown and James Heathers in 2016, the test is simple enough to run with pencil and paper or a spreadsheet, but it proved immediately powerful: when applied to papers in leading social psychology journals, they found inconsistencies in a substantial fraction of tested articles — most of which had passed peer review.

The test does not prove fabrication. An impossible mean could arise from a transcription error, a rounding mistake, or a copy-paste problem. But it proves that something in the paper is wrong — itself significant when the paper has influenced downstream work or clinical guidelines.

2. Methodological Overview

The core intuition is easiest to grasp with a small example. Suppose ten participants each rate their agreement on a scale from 1 to 7. The sum of their responses must be an integer between 10 and 70. The mean is that integer sum divided by 10 — so the mean can only take specific decimal values such that (mean $\times$ N) is a whole number.

More precisely: multiply the reported mean by the reported N. If the result is not (very close to) a whole number, the mean is impossible. The test formalises this into a single check:

$$(\text{Reported Mean} \times N) \bmod 1 \approx 0$$

Where mod 1 gives the fractional part. If it is not negligibly close to zero (or to one, to allow for rounding), the mean cannot exist for integer data with that sample size.

In practice, you need to account for rounding: a reported mean of 3.47 could represent a true mean anywhere between 3.465 and 3.475. The test checks whether any integer sum within that rounding window is achievable. If no integer multiple falls within the window, the mean is impossible regardless of rounding. If one does, the mean is GRIM-consistent.

The test extends naturally to standard deviations (GRIMMER, Allard 2018) and to subgroup means, percentages, and proportions. Each extension applies the same core logic: what integer sums are possible given N, and does the reported statistic correspond to one of them?

3. Example Application: Wansink Food Lab

Brian Wansink's Cornell Food and Brand Lab produced dozens of widely-cited findings about how environmental cues affect eating behavior. Following concerns raised in 2016, James Heathers and Nicholas Brown applied GRIM systematically to the lab's published papers.

One finding: a paper reported that children who saw an Elmo sticker on an apple chose the apple 71.7% of the time in a sample of N = 23. Multiply: $0.717 \times 23 = 16.491$. This is not a whole number — no arrangement of 23 children can produce a mean of 0.717. The reported percentage is impossible.

Across the Wansink papers, Heathers and Brown identified dozens of GRIM-inconsistent means. When SPRITE was applied alongside GRIM, the pattern was too systematic to be explained by transcription error — including one finding implying a child ate approximately 60 carrots at a single sitting.

Cornell University eventually retracted 13 of Wansink's papers following a formal misconduct investigation. The GRIM test was the statistical tripwire that initiated the scrutiny.

4. Requirements & Applicability

The GRIM test is not universally applicable. Its usefulness depends on properties of the data and what the paper actually reports.

Sample size (N)	Must be reported. Most powerful when $N < 60$ per group — with larger N , more decimal values become reachable and the test rarely fires.
Scale type	The underlying response scale must be integer-valued: Likert scales, binary 0/1 variables, counts, ordinal scores. Continuous measurements cannot be tested.
Decimal precision	The mean must be reported with at least one decimal place. A mean reported as an integer is GRIM-consistent with any N — the test only gains power when rounding is observable.
Reported statistic	Means or proportions are the primary target. Does not apply directly to SDs, correlations, or p-values — though GRIMMER and SPRITE extend the logic.
Raw data required?	No. The test operates entirely on published summary statistics — the reported mean and N . This is what makes it feasible as a post-publication check by anyone with access to the paper.
Scale bounds	Knowing the minimum and maximum response value allows an additional bounds check: the mean must fall within [scale min, scale max]. Important for proportion-based tests.

5. Limitations

- Cannot detect fabrication in continuous data (reaction times, physiological measurements). The majority of findings in many domains are not testable.
- A GRIM-consistent result is not validated — it simply cannot be ruled out. Many errors and most fabrications will not produce GRIM-inconsistent means.
- The test loses power as N grows. For $N > 100$, almost any decimal mean is achievable and inconsistencies are rarely flagged.
- Weighted means, trimmed means, or means computed after data transformation may appear GRIM-inconsistent without being wrong. Context matters.
- Requires knowing the response scale. Papers that do not report the scale, or that aggregate across multiple differently-scaled items, may produce apparent inconsistencies that are artefacts of aggregation.

6. How to Apply

Manual / spreadsheet method

- Identify every mean (or proportion) reported in the paper along with its associated N .
- Multiply each mean by its N to get the implied integer sum.
- Check whether the implied sum is very close to a whole number — compute $(\text{mean} \times N)$ and check whether the fractional part is < 0.001 or > 0.999 .
- Account for rounding: check whether any integer within the rounding window ($\text{mean} \pm 0.005$ for two-decimal reporting) produces a whole number sum when multiplied by N .
- Flag any means where no integer sum is reachable even after allowing for rounding.

Web-based tool

The GRIM app (grimapp.site, maintained by Heathers and Brown) accepts a mean, N, and scale bounds and returns an immediate consistency verdict with a visualization of which integer sums are reachable.

R / Python

The grimmer R package (Anaya 2016) implements both GRIM and GRIMMER for batch processing. It accepts a data frame of (mean, N, item_count, scale_min, scale_max) tuples and returns a consistency flag for each.

7. Related Tools

SPRITE	Extends GRIM: instead of asking 'is this mean possible?', SPRITE asks 'what distributions of integer responses could produce this mean and standard deviation together?' Reveals when the combination of M and SD implies a physically impossible or implausibly narrow data distribution.
statcheck	Automated extraction and recalculation of p-values from PDFs. Operates on test statistics rather than means; complements GRIM by catching a different class of reporting error.
GRIMMER	Extends GRIM to standard deviations: checks whether the reported SD is achievable given M, N, and integer constraints. Substantially more demanding mathematically but follows the same core logic.

Key Citations

- Brown, N. J. L. & Heathers, J. A. J. (2017). The GRIM test: A simple technique detects numerous anomalies in the reporting of results in psychology. *Social Psychological and Personality Science*, 8(4), 363–369. <https://doi.org/10.1177/1948550616673876>
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- Anaya, J. (2016). The GRIMMER test: A method for testing the validity of reported measures of variability. *PeerJ Preprints*. <https://doi.org/10.7287/peerj.preprints.2400v1>
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